

**Mathematical Foundations for
Machine Learning - Mid-Semester 1
Mock 01 Solutions**

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MFML Midsem 1 - Mock 01 Answer Key

Q1 - Linear system classification

Governing rule and shapes. Here A is 2 by 2, x is 2 by 1, and b is 2 by 1. A system is consistent exactly when $\text{rank}(A)=\text{rank}([A|b])$; if consistent, it is unique exactly when that common rank equals the number n of unknowns.

Substitution and intermediate work. The second coefficient row is twice the first, and the augmented row $(2,2|6)$ is twice $(1,1|3)$. Hence both matrices have one nonzero independent row:

$$\text{rank}(A) = 1, \text{rank}([A|b]) = 1, n = 2$$

Conditions and interpretation. Equal ranks make the system consistent. The common rank is below n , so one free real degree of freedom gives infinitely many calibrations; for example, setting $y=t$ gives $x=3-t$. The two equations supply the same information, so they cannot select one calibration. A further equation whose coefficient row is not proportional to $(1,1)$ would add an independent constraint and, if consistent, raise the coefficient rank to 2, selecting one pair.

Q2 - Vector and affine spaces

This is not a calculation, so formula, substitution, and numerical interpretation requirements are not applicable. A finite list is linearly independent when only the all-zero coefficients make its linear combination zero. It spans a space when every vector in that space is a linear combination of the list. It is a basis exactly when both properties hold. A subspace U contains zero and is closed under addition and scalar multiplication. An affine set $a+U=\{a+u:u \text{ in } U\}$ is a shifted copy of U and need not contain zero. It is a subspace precisely when it contains zero, equivalently when a belongs to U , in which case $a+U=U$.

Q3 - Projection

Governing formula and condition. For a unit column q , the projection is $p=q(q^T x)$. The shortcut requires $q^T q=1$.

Substitution and intermediate work. First,

$$q^T q = (1/\sqrt{2})^2 + (1/\sqrt{2})^2 = 1$$

$$q^T x = (3+4)/\sqrt{2} = 7/\sqrt{2}$$

Both q and x are 2 by 1, while $q^T x$ is scalar. Therefore

$$p = q(q^T x) = \begin{bmatrix} 7/2 \\ 7/2 \end{bmatrix} \quad r = x-p = \begin{bmatrix} -1/2 \\ 1/2 \end{bmatrix}$$

Check and interpretation. $q^T r = (-1/2+1/2)/\sqrt{2}=0$, so the residual is orthogonal to the projection direction. The projection is the component of x along $\text{span}(q)$; the residual is the perpendicular component left after removing it.

Q4 - Eigenpairs and Cholesky

Definition and existence condition. An eigenpair satisfies $Av = \lambda v$, where v is nonzero. Ordinary real Cholesky requires a real symmetric positive-definite square matrix and returns a lower-triangular factor with positive diagonal.

Governing matrix equation. Let

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \quad L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix}$$

Matching $LL^T = A$ gives the entry equations below.

$$\begin{aligned} l_{11}^2 &= 4 \\ l_{11}l_{21} &= 2 \\ l_{21}^2 + l_{22}^2 &= 2 \end{aligned}$$

Substitution and intermediate work. Positive diagonal entries give $l_{11}=2$, then $l_{21}=1$, then $1+l_{22}^2=2$, so $l_{22}=1$.

$$L = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad LL^T = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} = A$$

Conditions and interpretation. A is real and symmetric. The solved L has positive diagonal, so it is nonsingular. For any nonzero real column z :

$$z^T A z = z^T L L^T z = \|L^T z\|_2^2 > 0$$

z is nonzero and L is nonsingular, so $L^T z$ is nonzero

The squared norm is positive because nonsingular L^T cannot send nonzero z to zero. Thus A is positive definite, ordinary real Cholesky is valid, and the reconstruction checks every entry. In the covariance interpretation, the real nonsingular factor can generate the covariance as LL^T ; there is no zero-variance direction.

Q5 - Diagonalization and SVD

Conditions and proof. A square matrix is diagonalizable exactly when it has a full eigenvector basis. Distinct eigenvalues have independent eigenvectors, so statement (i) is true. Invertibility instead requires every eigenvalue to be nonzero, so statement (ii) is false.

Counterexample. One valid construction is

$$D = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

Its distinct eigenvalues 0 and 2 make its coordinate eigenvectors independent, so it is diagonalizable. Its zero eigenvalue gives $\det(D)=0$, so it is singular.

Low-rank calculation and interpretation. If singular values are ordered $\sigma_1 \geq \sigma_2 \geq \sigma_3$, Eckart-Young gives

$$\|A - A_2\|_2 = \sigma_3 = 1$$

The best rank-at-most-2 approximation therefore has spectral-norm error 1, the largest discarded stretch factor. Since $1 \leq 1.5$, the compression meets the allowed tolerance; its worst-direction change is at most one unit.

Q6 - Jacobian interpretation

Governing convention and shapes. In the output-by-input convention, entry (i,j) is $\partial g_i / \partial x_j$. Since $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, its Jacobian is 2 by 2, and it multiplies the 2 by 1 displacement h .

Derivatives and substitution. Componentwise differentiation gives

$$Jg(x,y) = \begin{bmatrix} 2x & 1 \\ y & x \end{bmatrix} \quad Jg(2,3) = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$$

Intermediate multiplication. With $h=[0.01, -0.02]^T$,

$$Jg(2,3)h = \begin{bmatrix} 4(0.01)+1(-0.02) \\ 3(0.01)+2(-0.02) \end{bmatrix} = \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix}$$

Interpretation and condition. The first output rises by about 0.02 and the second falls by about 0.01. This is a local first-order estimate, accurate for sufficiently small displacements when g is differentiable nearby.

Q7 - Backpropagation and automatic differentiation

Forward substitution. The graph contains scalars, so all local derivatives and the final seed are scalar.

$$x=2 \rightarrow u=x^2=4 \rightarrow y=3u+1=13 \rightarrow L=y^2=169$$

Reverse formula and intermediate work. Seed the scalar output with $dL/dL=1$, then multiply local derivatives backward.

$$dL/dy = 2y = 26$$

$$dL/du = (dL/dy)(dy/du) = 26(3) = 78$$

$$dL/dx = (dL/du)(du/dx) = 78(4) = 312$$

Interpretation. The loss changes locally at rate 312 per unit change in x . This is automatic differentiation because exact local derivative rules for executed operations are composed on the recorded graph; no symbolic expression expansion or finite input perturbation is used.

Q8 - Hessian and optimization conclusion

Gradient and stationary point. Differentiate and set both components to zero:

$$\text{grad } f(x,y) = \begin{bmatrix} 2x-4 \\ 4y+8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{so } (x,y)=(2,-2)$$

Hessian condition and local interpretation.

$$H = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad z^T H z = 2z_1^2 + 4z_2^2 > 0 \text{ for every nonzero } z$$

Thus H is positive definite and the stationary point is a strict local minimum.

Global check and interpretation. Completing squares gives

$$f(x,y) = (x-2)^2 + 2(y+2)^2 - 12$$

Both squared terms are nonnegative on all of $\mathbb{R}^2$, so $(2,-2)$ is the unique global minimum and the minimum value is -12 .

MFML Midsem 1 - Mock 01 Point Rubric

Award each one-mark point independently. Accept an equivalent valid method. Apply arithmetic follow-through when later work is correct from an earlier arithmetic slip and required conditions remain valid.

Q1

- Q1-P1 - 1 mark, Easy, Session 1, numerical reasoning: Governing rank rules and 2 by 2, 2 by 1 shapes are stated.
- Q1-P2 - 1 mark, Standard, Session 1, numerical reasoning: Row dependence is calculated, both ranks are obtained, and the solution set is classified by substitution into the rank rule.
- Q1-P3 - 1 mark, Standard, Session 1, scenario analysis or interpretation: The analyst's nonunique calibration is interpreted as one free real degree of freedom, with an illustrative family or equivalent explanation.

- Q1-P4 - 1 mark, Standard, Session 1, scenario analysis or interpretation: An additional equation independent of the first is recommended, and the response explains why it would select one calibration.

Q2

- Q2-P1 - 1 mark, Easy, Session 2, property or definition: Linear independence is defined correctly.
- Q2-P2 - 1 mark, Easy, Session 2, property or definition: Spanning and the combined basis condition are correct.
- Q2-P3 - 1 mark, Standard, Session 2, property or definition: Subspace closure and zero requirements are complete.
- Q2-P4 - 1 mark, Standard, Session 2, comparison or distinction: The affine shift and exact condition for containing zero or being a subspace are compared. Calculation-answer components are not applicable.

Q3

- Q3-P1 - 1 mark, Easy, Session 3, numerical reasoning: Projection formula, unit-vector condition, and compatible shapes are stated.
- Q3-P2 - 1 mark, Standard, Session 3, numerical reasoning: Unit length and scalar coefficient are calculated with substitutions.
- Q3-P3 - 1 mark, Standard, Session 3, numerical reasoning: Projection and residual are obtained through visible intermediate work.
- Q3-P4 - 1 mark, Stretch, Session 3, scenario analysis or interpretation: The projection is interpreted as the component along the line and the residual as its perpendicular remainder; the zero dot-product check supports that interpretation.

Q4

- Q4-P1 - 1 mark, Easy, Session 4, property or definition: Eigenpair equation and nonzero-vector condition are complete.
- Q4-P2 - 1 mark, Standard, Session 4, property or definition: Cholesky formula, matrix shapes, positive-diagonal convention, and real-symmetric-positive-definite condition are stated.
- Q4-P3 - 1 mark, Standard, Session 4, numerical reasoning: Entry equations are formed and solved with substitutions and intermediate values.
- Q4-P4 - 1 mark, Stretch, Session 4, scenario analysis or interpretation: Reconstruction and positive definiteness support a decision that real Cholesky is valid, interpreted as a usable real covariance factor.

Q5

- Q5-P1 - 1 mark, Easy, Session 5, comparison or distinction: The full-eigenbasis condition for diagonalizability is stated.
- Q5-P2 - 1 mark, Standard, Session 5, comparison or distinction: The distinct-eigenvalue claim is proved using eigenvector independence.

- Q5-P3 - 1 mark, Standard, Session 5, comparison or distinction: A valid 2 by 2 counterexample separates invertibility from diagonalizability and states the nonzero-eigenvalue condition.
- Q5-P4 - 1 mark, Stretch, Session 5, scenario analysis or interpretation: The Eckart-Young calculation is used to decide that error 1 meets the 1.5 tolerance, with the discarded-direction meaning interpreted.

Q6

- Q6-P1 - 1 mark, Easy, Session 6, numerical reasoning: Component derivatives, output-by-input convention, and 2 by 2 Jacobian shape are correct.
- Q6-P2 - 1 mark, Standard, Session 6, numerical reasoning: Evaluation at (2,3) is substituted into the Jacobian.
- Q6-P3 - 1 mark, Standard, Session 6, numerical reasoning: Matrix-vector multiplication shows both intermediate component calculations.
- Q6-P4 - 1 mark, Stretch, Session 6, scenario analysis or interpretation: Both changes are interpreted with the differentiable, sufficiently-small-displacement condition.

Q7

- Q7-P1 - 1 mark, Easy, Session 7, numerical reasoning: All forward values and scalar shapes are calculated.
- Q7-P2 - 1 mark, Standard, Session 7, numerical reasoning: Scalar seed, local derivative formulas, intermediate reverse values, and final derivative 312 are correct.
- Q7-P3 - 1 mark, Stretch, Session 7, scenario analysis or interpretation: The result is interpreted and AD is distinguished from both symbolic differentiation and finite differences.

Q8

- Q8-P1 - 1 mark, Easy, Session 8, numerical reasoning: Gradient formula, zero substitution, and stationary point are calculated correctly.
- Q8-P2 - 1 mark, Standard, Session 8, scenario analysis or interpretation: Hessian, positive-definite condition, and strict-local-minimum interpretation are complete.
- Q8-P3 - 1 mark, Stretch, Session 8, scenario analysis or interpretation: Completing squares supplies the global condition, value, and unique-global-minimum interpretation.

Coverage totals

- Sessions 1-6: 4 marks each; Sessions 7-8: 3 marks each. No session exceeds 4 marks.
- Property or definition: 5 marks. Comparison or distinction: 4 marks. Numerical reasoning: 12 marks. Scenario analysis or interpretation: 9 marks. Each point's primary form follows its assessed action: computation is numerical, while scenario points require a decision, contextual meaning, or geometric interpretation.
- Numerical plus applied: 21 of 30 marks. Difficulty: Easy 9, Standard 15, Stretch 6 marks.
- Largest-remainder record for the standard paper: 30 one-mark units times target shares 0.30,0.50,0.20 gives quotas 9.00,15.00,6.00; floors are 9,15,6; remainders are all 0.00; no residual unit or tie-break is needed; final allocation is 9,15,6.